

NIKHIL POLEPALLI

SOFTWARE ENGINEER

New York City, New York | (502) 424-7440 | nikhil.s.polepalli@gmail.com | [LinkedIn](#)

PROFILE

Software engineer with 3+ years building scalable, cloud-native platforms and **distributed systems** at Capital One. Proven track record **owning complex features end-to-end**, architecting **high-volume asynchronous processing systems**, and optimizing **real-time data pipelines**. Expert in Python, Go, Java, and AWS with deep experience in **search optimization**, Kubernetes, and **generative AI tooling** for developer productivity.

EXPERIENCE

Senior Associate Software Engineer, Capital One – New York, NY

Jan 2026 - Present

- Reduced platform downtime by **75%** as measured by eliminating service interruptions, by owning multi-region failover infrastructure with **traffic routing**, **load balancing**, and enhanced **observability** using **Prometheus-based** monitoring and automated alerting systems.
- Delivered a high-volume, **consumer-facing** transaction platform for 4 payment networks as measured by **2M+ transactions daily** with **99.9% uptime**, by architecting **scalable web application** infrastructure with asynchronous processing and event streaming.
- Accelerated developer productivity by **35%** as measured by faster deployment cycles across 6 microservices, by implementing CI/CD automation and **full-stack** development tooling to streamline code generation and **cross-platform** deployment workflows.

Associate Software Engineer, Capital One – McLean, Virginia

Aug 2024 - Dec 2025

- Engineered **real-time data flows** as measured by processing **40M+ records/day** with full audit traceability, by designing **asynchronous job orchestration systems** using AWS Lambda, SQS, and DynamoDB.
- Drove **\$500K in annual cost savings** as measured by **50% runtime improvement**, by optimizing **data ingestion pipelines** and query performance using Python and Java, enabling faster **search and retrieval** across large-scale datasets.
- Improved system reliability by **40%** as measured by **5M+ daily API requests** served seamlessly, by building RESTful APIs powering **large-scale web applications** with evolvable data models and real-time notification systems for **end-to-end user experiences**.

Software Engineer, Chalkline Sports – Nashville, Tennessee

Jan 2024 - Aug 2024

- Improved **consumer-facing** platform performance as measured by **30% faster response times** and **99.9% uptime** under **10K+ concurrent users**, by optimizing database queries and implementing search optimization to enhance **user experience at scale**.
- Owned **end-to-end delivery** of product integrations as measured by 4 new clients and **15% user engagement increase**, by building **full-stack features** with RESTful APIs enabling automated reporting and real-time data exchange across **web platforms**.

Software Engineering Intern, Capital One – McLean, Virginia

Jun 2023 - Aug 2023

- Built **3 high-performance RESTful API endpoints** in Go as measured by **90% unit test coverage**, by **independently owning** API design and data models that improved developer onboarding and long-term maintainability.
- Created **React-based web application** as measured by **3x usability improvement** for **10K+ configurations**, by building **full-stack features** with search, filtering, and intuitive **UI/UX** capabilities supporting cross-team workflows.

Software Engineering Intern, Northrop Grumman – McLean, Virginia

Jun 2022 - Aug 2022

- Developed Python ETL script using pandas and SQL, automating manual data extraction and transformation workflows, and reducing processing time from hours to minutes through batch processing and data pipeline optimization.
- Engineered automated data processing system in Python, replacing manual ETL workflows with scheduled scripts that process 500k+ records/day, improving data quality and reducing operational overhead.

TECHNICAL SKILLS & CERTIFICATIONS

Certifications: AWS Certified Solutions Architect Associate

Languages: Python, Java, TypeScript, **JavaScript**, Go (Golang), SQL, C++

Frontend & Web: **React**, **Next.js**, HTML/CSS, **Responsive Web Design**

Backend & APIs: RESTful APIs, Kafka, Event-Driven Architecture

Databases: PostgreSQL, MySQL, DynamoDB, Redis, AWS S3

Cloud & Infrastructure: AWS (Lambda, SQS, ECS, EC2, CloudWatch), Docker, Kubernetes

Testing: JUnit, Mockito, WireMock, **End-to-End Testing**

EDUCATION

Vanderbilt University

Aug 2020 - May 2024

B.S. in Computer Science, Economics - GPA: 3.85